

Addendum: the correlation theorem behind the overlap integral, and sources for the pulse-area section

1 Equation (42) in full

The step in question is

$$O(\mathbf{d}) = \int u(|\mathbf{r}|) u(|\mathbf{r} - \mathbf{d}|) d^2r = \frac{1}{(2\pi)^2} \int |\tilde{u}(q)|^2 e^{i\mathbf{q} \cdot \mathbf{d}} d^2q. \quad (1)$$

It was quoted as "the correlation theorem". Written out it is five lines, and three of them are the ones that actually do work: the choice of Fourier convention, the symmetry of u , and the interchange of the two integrations.

Convention. The factor $1/(2\pi)^2$ in Eq. (1) is not universal – it is fixed by where one puts it in the transform pair. Throughout,

$$\tilde{u}(\mathbf{q}) = \int u(\mathbf{r}) e^{-i\mathbf{q} \cdot \mathbf{r}} d^2r, \quad u(\mathbf{r}) = \frac{1}{(2\pi)^2} \int \tilde{u}(\mathbf{q}) e^{+i\mathbf{q} \cdot \mathbf{r}} d^2q, \quad (2)$$

i.e. the whole $1/(2\pi)^2$ sits in the back transform [1, Ch. 2]. This is the same convention used for $\tilde{u}(q) = (4\pi/k^2) \Theta(k - q)$ in step (i), so the two are consistent.

Step 1: replace the shifted factor by its transform. Only the second factor is expanded; the first is left alone.

$$u(\mathbf{r} - \mathbf{d}) = \frac{1}{(2\pi)^2} \int \tilde{u}(\mathbf{q}) e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{d})} d^2q = \frac{1}{(2\pi)^2} \int \tilde{u}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}} e^{-i\mathbf{q} \cdot \mathbf{d}} d^2q. \quad (3)$$

The shift has become a pure phase factor $e^{-i\mathbf{q} \cdot \mathbf{d}}$ – this is the only property of the Fourier transform the derivation needs.

Step 2: insert and exchange the order of integration.

$$O(\mathbf{d}) = \int d^2r u(\mathbf{r}) \frac{1}{(2\pi)^2} \int d^2q \tilde{u}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}} e^{-i\mathbf{q} \cdot \mathbf{d}} \quad (4)$$

$$= \frac{1}{(2\pi)^2} \int d^2q \tilde{u}(\mathbf{q}) e^{-i\mathbf{q} \cdot \mathbf{d}} \underbrace{\int d^2r u(\mathbf{r}) e^{+i\mathbf{q} \cdot \mathbf{r}}}_{= \tilde{u}(-\mathbf{q})}. \quad (5)$$

The inner integral is the defining transform (2) evaluated at $-\mathbf{q}$, because the sign in the exponent is the opposite one. Interchanging the two integrations is legitimate here: u is bounded and square integrable and $\tilde{u}$ has compact support, so the double integral converges absolutely and Fubini's theorem applies. (For a Gaussian mode the same statement is immediate; for the Airy mode it is the compact support of $\tilde{u}$ that saves it, since u itself decays only as $r^{-3/2}$.)

Step 3: the symmetry of the mode. Two properties are needed, and the mode has both:

$$u \text{ real} \implies \tilde{u}(-\mathbf{q}) = \tilde{u}^*(\mathbf{q}) \quad (\text{Hermitian symmetry}), \quad (6)$$

$$u \text{ even, } u(-\mathbf{r}) = u(\mathbf{r}) \implies \tilde{u}(-\mathbf{q}) = \tilde{u}(\mathbf{q}) \quad (\text{substitute } \mathbf{r} \rightarrow -\mathbf{r}). \quad (7)$$

A radially symmetric mode is automatically even, and both the Gaussian and $2J_1(kr)/(kr)$ are real. Taken together, $\tilde{u}$ is itself real and even – as the explicit result $\tilde{u} = (4\pi/k^2)\Theta(k-q)$ shows. Therefore

$$\tilde{u}(\mathbf{q}) \tilde{u}(-\mathbf{q}) = \tilde{u}(\mathbf{q}) \tilde{u}^*(\mathbf{q}) = |\tilde{u}(\mathbf{q})|^2, \quad (8)$$

and Eq. (5) becomes

$$O(\mathbf{d}) = \frac{1}{(2\pi)^2} \int |\tilde{u}(\mathbf{q})|^2 e^{-i\mathbf{q} \cdot \mathbf{d}} d^2q. \quad (9)$$

Step 4: the sign in the exponent does not matter. Equation (9) carries $e^{-i\mathbf{q} \cdot \mathbf{d}}$ while Eq. (1) was written with $e^{+i\mathbf{q} \cdot \mathbf{d}}$. Both are right: $|\tilde{u}|^2$ is even in $\mathbf{q}$, so the substitution $\mathbf{q} \rightarrow -\mathbf{q}$ leaves the measure and the weight unchanged and only flips the exponent. Equivalently: O is the transform of a real even function and is therefore itself real and even, $O(-\mathbf{d}) = O(\mathbf{d})$ – which is also physically obvious, since the overlap of two spots cannot depend on which one is called the first.

Why "correlation" and "convolution" coincide here. In general the two differ,

$$(u \star v)(\mathbf{d}) = \int u(\mathbf{r}) v(\mathbf{r} - \mathbf{d}) d^2r, \quad (u * v)(\mathbf{d}) = \int u(\mathbf{r}) v(\mathbf{d} - \mathbf{r}) d^2r, \quad (10)$$

and only the convolution turns into a plain product of transforms. For an *even* v one has $v(\mathbf{d} - \mathbf{r}) = v(\mathbf{r} - \mathbf{d})$ and the two integrals are identical [1, Ch. 2]. That is the one line the phrase "since u is real and even the correlation equals the convolution" stands for.

Step 5: the check at $\mathbf{d} = 0$. Setting $\mathbf{d} = 0$ in Eq. (1) reduces it to Parseval's theorem in this convention,

$$O(0) = \int u^2 d^2r = \frac{1}{(2\pi)^2} \int |\tilde{u}|^2 d^2q = P_1, \quad (11)$$

so the normalisation of Eq. (1) is fixed by a quantity already computed independently in step (a). This is the cheapest available test of the prefactor and it passes.

Evaluation for the Airy mode. Now the special structure enters. An indicator function is idempotent, $\Theta^2 = \Theta$, and that is what makes the Airy case as easy as it is. With $\tilde{u}(q) = (4\pi/k^2)\Theta(k-q)$,

$$O^A(d) = \frac{1}{(2\pi)^2} \left(\frac{4\pi}{k^2} \right)^2 \int_{|\mathbf{q}| < k} e^{i\mathbf{q} \cdot \mathbf{d}} d^2q \quad \text{Eq. (9), } \Theta^2 = \Theta \quad (12)$$

$$= \frac{1}{4\pi^2} \cdot \frac{16\pi^2}{k^4} \cdot \frac{2\pi k J_1(kd)}{d} \quad \text{disc integral, step (ii)} \quad (13)$$

$$= \frac{4}{k^4} \cdot \frac{2\pi k J_1(kd)}{d} = \frac{8\pi J_1(kd)}{k^3 d} \quad (14)$$

$$= \underbrace{\frac{4\pi}{k^2}}_{P_1^A} \cdot \underbrace{\frac{2J_1(kd)}{kd}}_{u(d)}. \quad (15)$$

$$\boxed{O^A(d) = P_1^A u(d)} \quad (16)$$

The autocorrelation of the Airy mode is the mode itself. The reason is visible in the first line and nowhere else: the spectrum is flat inside the cutoff, so squaring it does nothing, and a function whose spectrum equals that of u up to a constant *is* u up to that constant. No such statement holds for the Gaussian, whose squared spectrum is narrower by $\sqrt{2}$ – which is exactly why $O^G(d) = P_1^G e^{-d^2/2w_0^2}$ carries $2w_0^2$ in the exponent instead of w_0^2 .

Numerical confirmation. On a 2048^2 grid over $40\,\mu\text{m}$ with $w_0 = 1.04\,\mu\text{m}$: $O(0)$ reproduces P_1 to machine precision (as it must, both being the same sum), the analytic $P_1^A = 4\pi/k^2$ is met to 1.2 %, and $O(d)$ agrees with $P_1^A u(d)$ to 1.1 % of the peak over $d = 0.2 \dots 6\,\mu\text{m}$ – the residual being the grid truncating the outer Airy rings, consistent with the 5 % truncation quoted in the main text. At the working-point separation $d = 3.305\,\mu\text{m}$ one gets $u(d) = +0.06308$.

2 Suggested citations for the pulse-area section

None of the derivations need a source; the starting points do. In the order they appear:

Place in the text	Citation
" $\int_0^\infty J_\nu^2(x) dx/x = 1/(2\nu)$ " (Airy spot power)	<code>\cite{dlmf}</code> – Weber–Schafheitlin type, DLMF §10.22
" $\int_0^\infty J_1(x) dx = 1$ " (prefactor of $\tilde{u}$)	<code>\cite{dlmf}</code>
" $\int_0^X J_0(t) t dt = X J_1(X)$ " (disc integral)	<code>\cite{dlmf}</code> ; equivalently one line from $\frac{d}{dz}[zJ_1(z)] = zJ_0(z)$
"the focal field is the Fourier transform of the pupil field" and the Airy form $2J_1(kr)/(kr)$	<code>\cite[Ch.~4,~6]{goodman2017}</code> , <code>\cite[Sec.~8.5]{bornwolf1999}</code>
the correlation theorem, Eq. (42), and Parseval	<code>\cite[Ch.~2]{goodman2017}</code>
"the excited states are eliminated adiabatically" and the two-photon Rabi frequency signed dipole matrix elements, hyperfine shifts, line centroids, lifetimes	<code>\cite{grimm2000}</code> for the light shift and scattering rate in the same far-detuned limit <code>\cite{sibalic2017}</code> (ARC), cross-checked against <code>\cite{steckrb85}</code>
the thermal width $\sigma^2 = \frac{\hbar}{2m\omega_r} \coth \frac{\hbar\omega_r}{2k_B T}$	<code>\cite{tuchendler2008}</code> (already in the appendix bibliography)

Two remarks on what *not* to cite. The boxed pulse-area formula and the cancellation of A_{eff} against $\tilde{P}_{\text{prof}}$ are bookkeeping carried out here in full – a reference would suggest the result is someone else’s. The same holds for the sinc: it is the Fourier transform of the integration window and is derived in three lines in the text.

References

- [1] Joseph W. Goodman. *Introduction to Fourier Optics*. W. H. Freeman / Macmillan Learning, New York, 4 edition, 2017. ISBN 978-1-319-11916-4. Fourier conventions, convolution and correlation theorems, Parseval’s theorem; band-limited imaging.

BibTeX source (additional entries)

These are new; tuchendler2008 is already in the appendix bibliography.

```
@book{goodman2017,  
  author   = {Goodman, Joseph W.},  
  title    = {Introduction to Fourier Optics},  
  edition  = {4}, publisher = {W. H. Freeman / Macmillan Learning},  
  address  = {New York}, year = {2017}, isbn = {978-1-319-11916-4},  
  note     = {Fourier conventions, convolution and correlation theorems,  
             Parseval's theorem; band-limited imaging}  
}  
  
@book{bornwolf1999,  
  author   = {Born, Max and Wolf, Emil},  
  title    = {Principles of Optics},  
  edition  = {7 (expanded)}, publisher = {Cambridge University Press},  
  address  = {Cambridge}, year = {1999}, isbn = {978-0-521-64222-4},  
  note     = {Sec.~8.5, the Airy pattern of a circular aperture}  
}  
  
@misc{dlmf,  
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  title     = {{NIST} Digital Library of Mathematical Functions,  
             Chapter-10 (Bessel Functions), \S10.22 (Integrals)},  
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  note      = {Release 1.2.4; accessed 17 September 2026}  
}  
  
@article{sibalic2017,  
  author = {\v{S}ibali\text{'}{c}, N. and Pritchard, J. D. and Adams, C. S. and  
           Weatherill, K. J.},  
  title  = {{ARC}: An open-source library for calculating properties of  
           alkali {R}ydberg atoms},  
  journal = {Computer Physics Communications},  
  volume = {220}, pages = {319--331}, year = {2017},  
  doi    = {10.1016/j.cpc.2017.06.015}  
}  
  
@article{grimm2000,  
  author = {Grimm, Rudolf and Weidem\text{''}{u}ller, Matthias and  
           Ovchinnikov, Yurii B.},  
  title  = {Optical Dipole Traps for Neutral Atoms},  
  journal = {Advances in Atomic, Molecular, and Optical Physics},  
  volume = {42}, pages = {95--170}, year = {2000},  
  doi    = {10.1016/S1049-250X(08)60186-X},  
  note   = {also \url{https://arxiv.org/abs/physics/9902072}}  
}  
  
@misc{steckrb85,  
  author    = {Steck, Daniel A.},  
  title     = {Rubidium 85 {D} Line Data},  
  howpublished = {\url{https://steck.us/alkalidata}},  
  note      = {accessed 17 September 2026}  
}
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